

STEPS IN PHARMACOECONOMIC EVALUATION

Steps in an Economic Evaluation

There are 8 basic steps to conduct PE evaluation

1. Establish the framework for the analysis
2. Determine the study design
3. Select the effectiveness measure and method for obtaining the data.
4. Identify relevant resources and measure usage, select unit prices, and estimate costs in each year.
5. Discount costs and effects in each year in the future and sum over all years in the time horizon for the study.

Cont...

6. Compute an incremental decision index
7. Conduct sensitivity and/or statistical analyses to determine the influence of uncertainty on conclusions
8. Interpret the cost-effectiveness results

1. Establishing the Framework

Decision Context

- Perspective
- Target Population
- Comparators

Study Boundaries

- Groups of Persons
- Types of Outcomes
- Time Horizon

Decision index

-out come measures (CEA, CUA and CBA)

Decision or Problem Context

- Determining the study question
 - clear study question
 - question -- economic significance
- Is there a specific problem which needs decision to be made? and if so, who is the decision maker?
- If there is no specific decision to be made, then what is the issue/problem/question (**economic dilemma**) that bears addressing?
- Who are the likely stakeholders and what are their economic issues?
- The context guides the choice of study *perspective, target population, and comparator*

Perspective

Common Perspectives for economic evaluation are

- Societal
- Government
- Third-party payer
- Institutional (Providers)
- Patient
- A pharmacoeconomic evaluation can assess **the value of a product or service** from **single or multiple perspectives**

Importance:

- Perspective affects the **scope of costs and outcomes** to be assessed

Perspective...

- **Patient Perspective**
 - paramount because patients are **the ultimate consumers** of healthcare services
- Costs--- what patients pay for a product or service
 - the portion not covered by insurance
- Consequences, from a patient's perspective, are the **clinical effects**, both positive and negative, of a program or treatment

Perspective...

- various costs might include **insurance copayments** and **out-of-pocket drug costs**, as well as indirect costs, such as lost wages
- This perspective -- considered when assessing the **impact of drug therapy on QOL** or
- if a patient will pay **out-of-pocket expenses** for a healthcare service

Perspective

- **Provider Perspective**

- Costs-- the actual expense of providing a product or service, regardless of what the provider charges.
- Providers can be hospitals or private-practice physicians.
- From this perspective,
- direct costs such as drugs, hospitalization, laboratory tests, supplies, and salaries of healthcare professionals can be identified, measured, and compared.

Perspective...

- However, indirect costs can be of less importance to the provider.
 - When making formulary management or drug use policy decisions,
 - the viewpoint of the healthcare organization should dominate.

Perspective...

- **Payer Perspective**

- Payers include insurance companies, employers, or the government.
- costs represent charges for healthcare products and services allowed or reimbursed by the payer.
- The primary cost for a payer is of a direct nature.

Perspective...

- indirect costs, such as
 - lost workdays (absenteeism),
 - being at work but not feeling well and therefore having lower productivity also can contribute to the total cost of healthcare to the payer
- the payer's perspective should be employed,
 - When insurance companies and employers
 - contract/select healthcare benefits for their employees

Perspective...

- **Societal Perspective**
 - the broadest of all perspectives
 - Theoretically, all direct and indirect costs are included in an economic evaluation
 - Costs
 - patient morbidity and mortality and the overall costs of giving and receiving medical care

Perspective...

- consequences
 - all the important consequences an individual could experience
- In countries with nationalized medicine, society is the predominant perspective

Target Population for the Treatment

- Consider subgroups based on expected differences in effectiveness, costs, preferences.
- subgroups based on
 - Age
 - Sex
 - Demographic
 - Economic status
 - Disease status

Comparator(s)

- Most commonly used alternative (usual care) and least expensive alternative
- Other alternatives may be relevant (e.g., most effective, “do nothing” alternative)

Study Boundaries/Scope

This is about making the **study do-able**.

Boundaries or limits specified for:

- Time horizon
- Groups of affected persons to be considered, **e.g., patients $\pm$ family**

Target popn-----→ groups of persons

- Types of **health and non health outcomes** to be included.

Identifying the appropriate pharmacoeconomic method

- The appropriate type of pharmacoeconomic method to be employed should be chosen
- cost-minimization analysis (CMA),
- cost-benefit analysis (CBA),
- cost-effectiveness analysis (CEA), and
- cost-utility analysis (CUA).

An Exercise in Framing the Analysis

- You are a hospital administrator of a 500 bed hospital. You were recently approached by the director of your hospital pharmacy with a proposal to establish a clinical pharmacist-managed anticoagulation clinic (AC) at your hospital. Presently, physicians from the ambulatory care and cardiology clinics refer patients to the anticoagulation service, which is staffed on a rotational basis by a resident in hematology. Your director of pharmacy feels that a clinical pharmacist-managed clinic will be superior to the present system.

Case Questions

1. What is the target population and group?
2. What would be the appropriate perspective for an economic evaluation of alternatives?
3. What alternatives/comparators should be considered?
4. Which type of consequences would it be most meaningful to measure?
5. Which economic evaluation technique should therefore be used?
6. What would be the decision index?

2. Determine the Study Design

- **Primary (prospective)** research design: piggyback (linked) PE study onto a Phase III clinical trial
- Primary research design: dedicated cost-effectiveness trial (“pragmatic” or simple, randomized trial)
- **Secondary (retrospective)** research design: uses existing clinical and administrative data, often from more than one source
- **Modeling design:** *A mathematical model of the sequence of clinical events after different treatment alternatives using existing data from multiple sources (published literature, unpublished clinical trial data, administrative databases and expert estimates) to estimate the consequences of each alternative*

NB: Be aware of the pros and cons of each design.

Sources of data

- Many sources of clinical evidence
 - the “**gold standard**” is normally considered to be the **randomized controlled trial (RCT)** or
 - **meta-analyses of RCTs**
- Randomized trials are excellent data sources on the effects of therapies, but
 - limited **applicability to a more general patient population**

RCTs+ Meta analyses

- Meta-analyses provide summary measures for parameters, but
 - studies considered are generally limited to randomized trials, thus **limiting generalizability**
- To use RCTs – the foll. should be considered
 - Are the potential results of sufficient importance to justify **the expense of RCT**

Observational studies

- **Incidence and prevalence** data may best come from
 - observational studies and large cohort
- Cohort studies -- useful for
 - **risk factor determination** and
 - determining the **natural history of an illness**

Observational studies...

- Observational studies -- should be considered for use as a primary approach only under certain conditions
- When;
 - the primary outcome is a **rare event**
 - **multiple comparators** are considered desirable
 - the **primary outcome has a long term horizon**

Observational studies...

- objective of a study is to evaluate **a single intervention**
 - cost of t/t before +after introduction of new therapy.
- the intervention in question has been **available for a long time**
- conducting RCT is **ethically questionable**
 - **magnitude of effect is very large** or
 - **breakthrough treatments of life-threatening** diseases

Observational studies...

- **randomization is not an acceptable option** for patients,
 - **surgery versus pharmacotherapy** or
 - **placebo- controlled trials in life-threatening illnesses.**
- **assessment of practice patterns** is part of evaluation.

Secondary Research Design: Retrospective Cohort

Involves use of existing data:

- Efficacy or effectiveness data from a prior clinical study
- Health care utilization data
 - Prior clinical study
 - Claims database
 - Hospital records
 - Disease registry

Retrospective data sources

- Administrative databases
 - for broad population-based estimates of **disease** and **effectiveness of therapies**
- Existing databases can provide **effectiveness and "real world" data**
- Retrospective analyses -- relatively **inexpensive** and can **be done quickly**

Retrospective data sources...

- **reflective of specific populations** that
 - cannot be easily studied using RCTs, and for which data may be difficult to obtain
- Relative to RCTs,
 - retrospective studies tend to **cover more realistic time frames**, and
 - **are not constrained** by the limitations of a set **trial period**

Retrospective data sources...

- Usually, the **sample sizes** of retrospective databases are much **larger than RCTs**,
 - enhancing their **statistical power** to detect important differences in outcomes
- Despite the advantages of retrospective database analyses, there are **challenges** when used for health economic evaluations

Retrospective data sources...

- validity of retrospective analyses is often questioned due to
 - the potential for **selection bias**,
 - **confounding** factors,
 - **data quality** and
 - **privacy issues**

Retrospective data sources...

- In addition-- there may be a **time lag** in the availability of information about **new therapies** + its **incorporation into and availability from databases**,
- the **timeliness of the economic evaluations** resulting from a retrospective analysis can be questioned

3. Measuring Effectiveness

3.1 Select Effectiveness Indicator

- Must be consistent with the treatment objective of the drug/program being evaluated
- Should incorporate both benefits and harms of alternative interventions
- Final health outcomes preferred to intermediate (surrogate) outcomes

3.2 Determine How Effectiveness Data will be Obtained

An economic evaluation is "*only as good as the effectiveness data it is built upon*".

Strength of evidence is a function of:

- Trial design
- Data sources—direct observation vs. self-report vs. database vs. experts
- Measurement methods-valid and reliable

What is validity and Reliability?

Study Questions on Effectiveness

1. Give example for the final outcome for each of the following:
 - Drug treatment for glaucoma
 - Medication management support program in a community pharmacy
2. If you had unlimited funding and time to do a pharmacoeconomic study, which type of design would you use and why?

4. Principles of Cost Estimation

- An economic cost is incurred whenever a resource is consumed. The value of that resource consumption is its opportunity cost, that is, the value of the benefits foregone because that resource is not available for its best alternative use.
- In perfectly functioning markets, market prices reflect opportunity cost; however, the health care market is considered to be imperfect. Prices need to be examined for extreme departures from opportunity cost.
- For “no cost” resources, e.g., volunteer time, space, need to impute an opportunity cost.

Steps in Cost Estimation

1. Identify Resources (inputs or consequences) Relevant to Study Perspective and Boundaries

□ Health care sector (direct medical costs)

- Out-of-pocket costs for health care
- Resources consumed in providing the intervention
- Health care used as a result of the intervention, e.g., to treat adverse events

□ Direct nonmedical:

- Out-of-pocket costs for resources needed to cope with the illness (e.g., home renovation, child care)
- Transport
- Time: family member/volunteer providing informal care and patient time to receive care

Cont...

❑ Indirect nonmedical costs :

- Productivity losses: loss in productive capacity (ability to work) **as a consequence** of the illness or death
 - Other sectors (external) - resources from other public agencies or the voluntary sector (e.g., social services)
- ❑ Intangible – i.e. the pain, worry or other distress which a patient or their family might suffer. These may be impossible to measure in monetary terms, but are sometimes captured in measures of quality of life.

2. Measure Resource Consumption

2.1 Determine Appropriate Level of Accuracy

Micro-Costing - detailed inventory, measurement, and valuation of **resources consumed** (inputs) in the production of a service or product, e.g., dispensing

- Human resource – technician time, pharmacist time
- Supplies container + label, drug info printouts
- Equipment_ computer + software, tablet counting machine
- Overhead costs (House rent, water, electricity, internet, etc)

Gross-Costing - count the number of **units of service consumed** (outputs) and apply a gross estimate of service cost, often what is typically charged,

e.g., - # Rx dispensed x dispensing fee
- MD visits x visit fee

2.2 Collect Resource Utilization Data

Quality of evidence = f (data source and instrumentation)

- Direct measurement, e.g., trial data collectors
- Archival data – health record, claims data
- Self-report–diary (prospective); survey or interview (retrospective)
- Expert estimate of services/treatments/tests provided to the average patient

Main types of cost

Total cost= sum of all the costs of producing a particular quantity of output

Average cost = total cost /quantity e.g. cost per patient, cost *per diem*

Marginal cost = the extra cost of producing one *extra* unit of output

Fixed costs = cost which do not vary with the quantity of output in the short run e.g. rent, equipment, lease payments, some wages and salaries

Variable costs = costs which vary with the level of output, e.g. food, medical/surgical supplies

3. Place a Value on Resources Consumed

- 3.1. Select a unit cost for each type of resource consumed
- A market price appropriate to the perspective of the analysis
 - For studies with a time horizon greater than 1 year, use unit costs from the base year, a concept called constant dollars (controls for price inflation)
 - For nonmarket goods and services, impute an opportunity cost
 - Consider inflation/deflation rate of overhead (shared) costs
 - Appropriateness of price selected depends on credibility of the source selected from the perspective of the analysis.

Valuing Resources cont'd

- 3.2 Calculate the total cost of each resource consumed in a given year = price per unit in constant dollars x units consumed
 - Special case of capital costs—spread over time
- 3.3 Calculate total annual costs by summing costs across all resources consumed
- 3.4 Discount costs incurred in each future year to their present value
- 3.5 Sum the stream of discounted costs over the time horizon of the study

5. Discounting Future Costs: Reasons, Why?

- Question: Should costs and benefits occurring at different points in time, be given equal weighting?
- If I offered to give you \$1000 today OR \$1000 in 5 years, which would you choose?
- Would you rather pay me \$1000 today OR \$1000 in 5 years?
- Generally individuals prefer to receive a benefit today and to incur a cost later
- Economists call this the notion of *time preference*
- Discounting is a process by which costs and benefits which occur at different times are made comparable by expressing future costs and benefits in terms of their present value

Cont...

- Human beings prefer to put off payment of costs into the future.
- Reasons are psychological and financial: frees up \$\$\$ for present consumption or investment
- Therefore a cost incurred in the future seems/is of less importance than the same cost incurred in the present, e.g., \$1000 after 1 year vs. now
- In order to reflect this lesser value of a deferred or future cost, we discount the future cost to a "present value", e.g., \$1000 becomes \$950 when viewed from the present

Mechanics of Discounting

Present value of a stream of costs over time

$$= \sum C_t / (1 + r)^t$$

$$= C_1 / (1 + r)^1 + C_2 / (1 + r)^2 + C_3 / (1 + r)^3 + \dots$$

Where C_t = costs at time t and r = discount rate

Discount Rate

- Historically 5%
- Newer guidelines 3% (U.S.), 6% (U.K.)

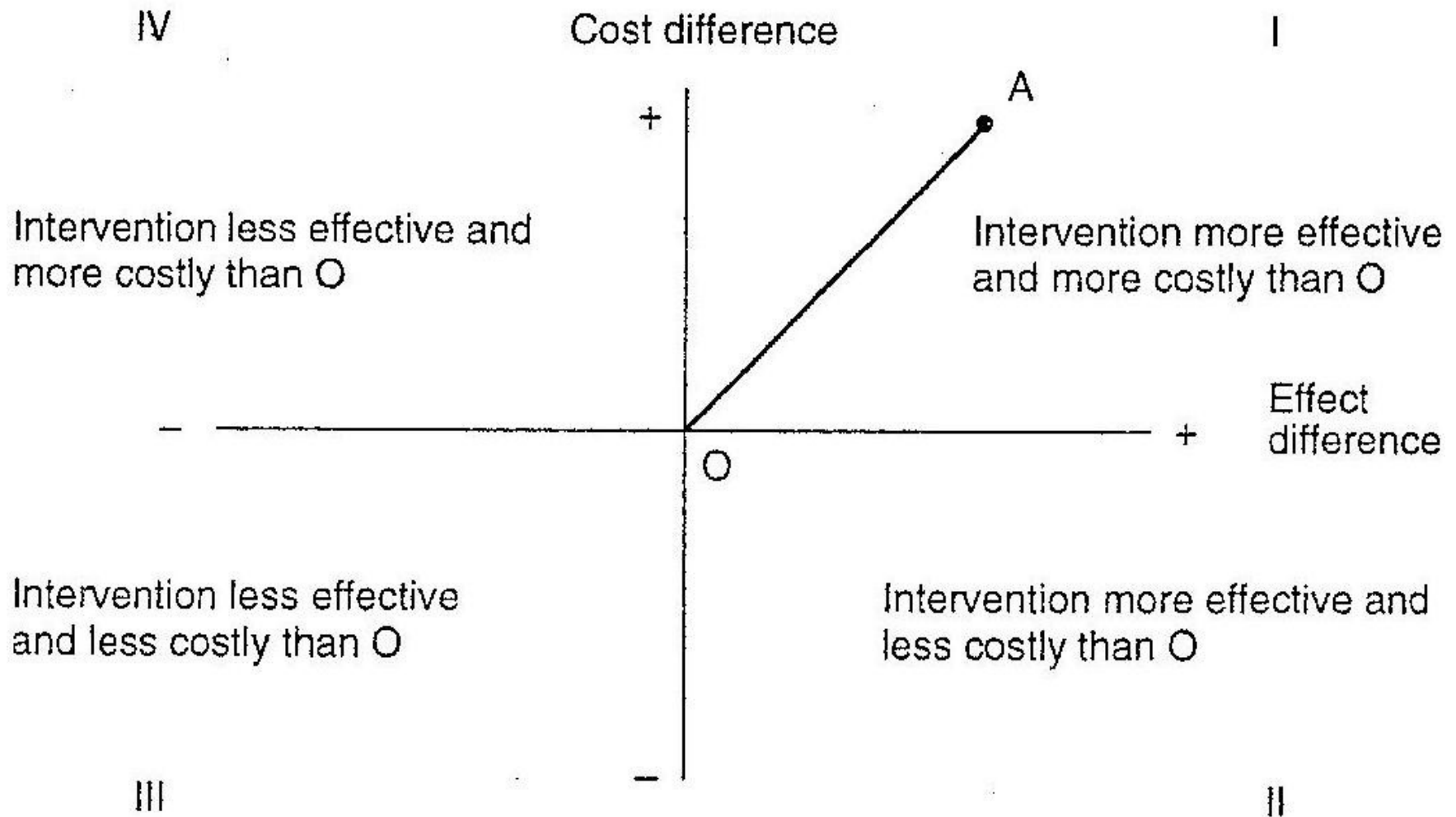
Example: Consider a cost of \$1000 incurred at end of years 1, 2, 3, & 25, where $r = 5\%$

Year	Future Cost	Present Value
1	\$1000	\$952
2	\$1000	\$907
3	\$1000	\$864
25	\$1000	\$295
Total Costs	\$4000	\$3018

Discounting Health Benefits

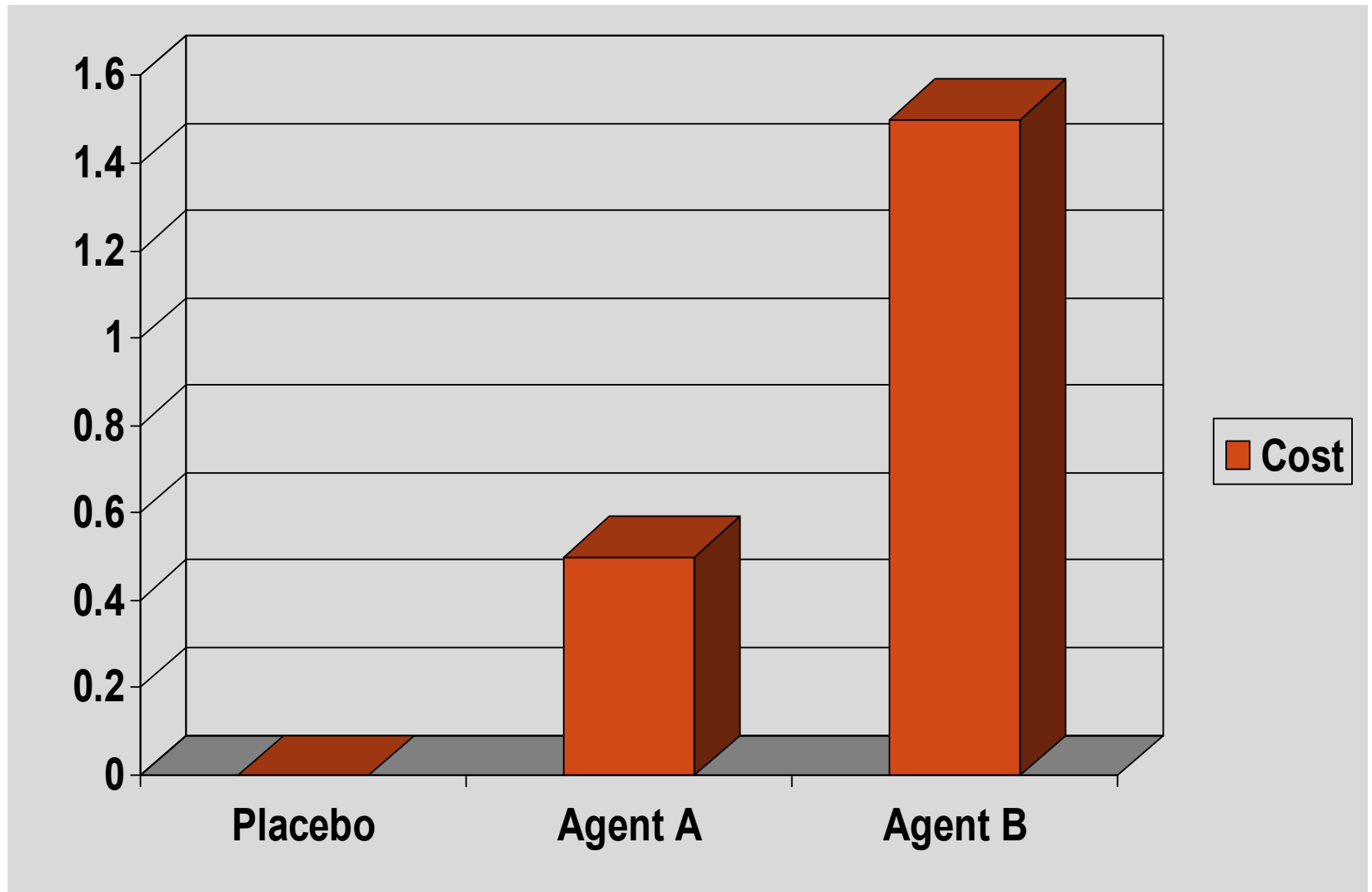
- To be consistent, we must discount health benefits incurring in the future as well.
- Should be at the same rate as for costs:
 - otherwise may become desirable to delay implementation indefinitely so that more income is earned, which would buy more lives saved
 - leads to resource allocation that is time neutral, i.e., cost-effectiveness does not differ for different time cohorts

6. Compute an incremental decision index

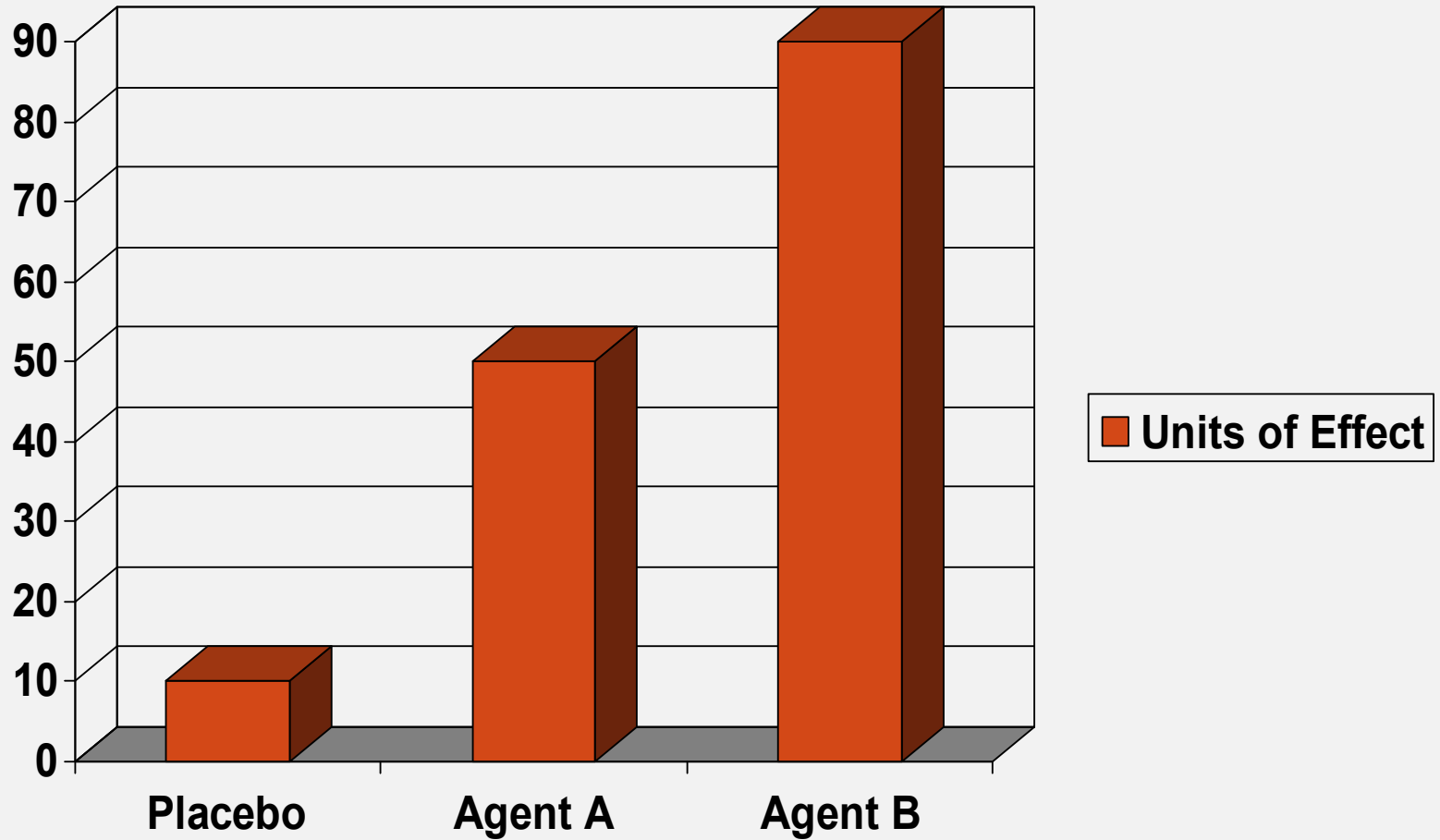


Adapted from Black (1990)

Incremental Cost Analysis (\$)



Incremental Effect Analysis



Conduct the Baseline (Base-Case) CE Analysis

1. Construct an incremental decision index/ratio
 - Check for dominance of one alternative over another, i.e., one alternative produces the same or greater effect and costs less OR provides more effect for the same cost
 - Eliminate dominated alternatives (“no brainers”)

2. Calculating incremental CE ratios (ICERs)

- Rank order remaining alternatives from lowest to highest total cost
- Calculate incremental ratio between least costly and next more costly alternative
 - $\text{Incremental costs}_{A-B} / \text{incremental health effect}_{A-B}$
- If there are more than 2 alternatives, proceed to next more costly alternative (C) and calculate an incremental ratio between B and C, and so on.

Why not calculate average ratios?

- Average $CE_A = \frac{\text{Total costs}_A}{\text{Total effects}_A}$
- Average $CE_B = \frac{\text{Total costs}_B}{\text{Total effects}_B}$
- Seems understandable that **the option with the lower ratio would be preferred.**
- **This is not always recommended especially if there is no shortage of resources**
- Average ratios don't answer the relevant economic question, which has to do with making decisions at the MARGIN

Incremental vs. Average Ratios

Source: Drummond, Sculpher, Torrance, O'Brien, Stoddart, 2008

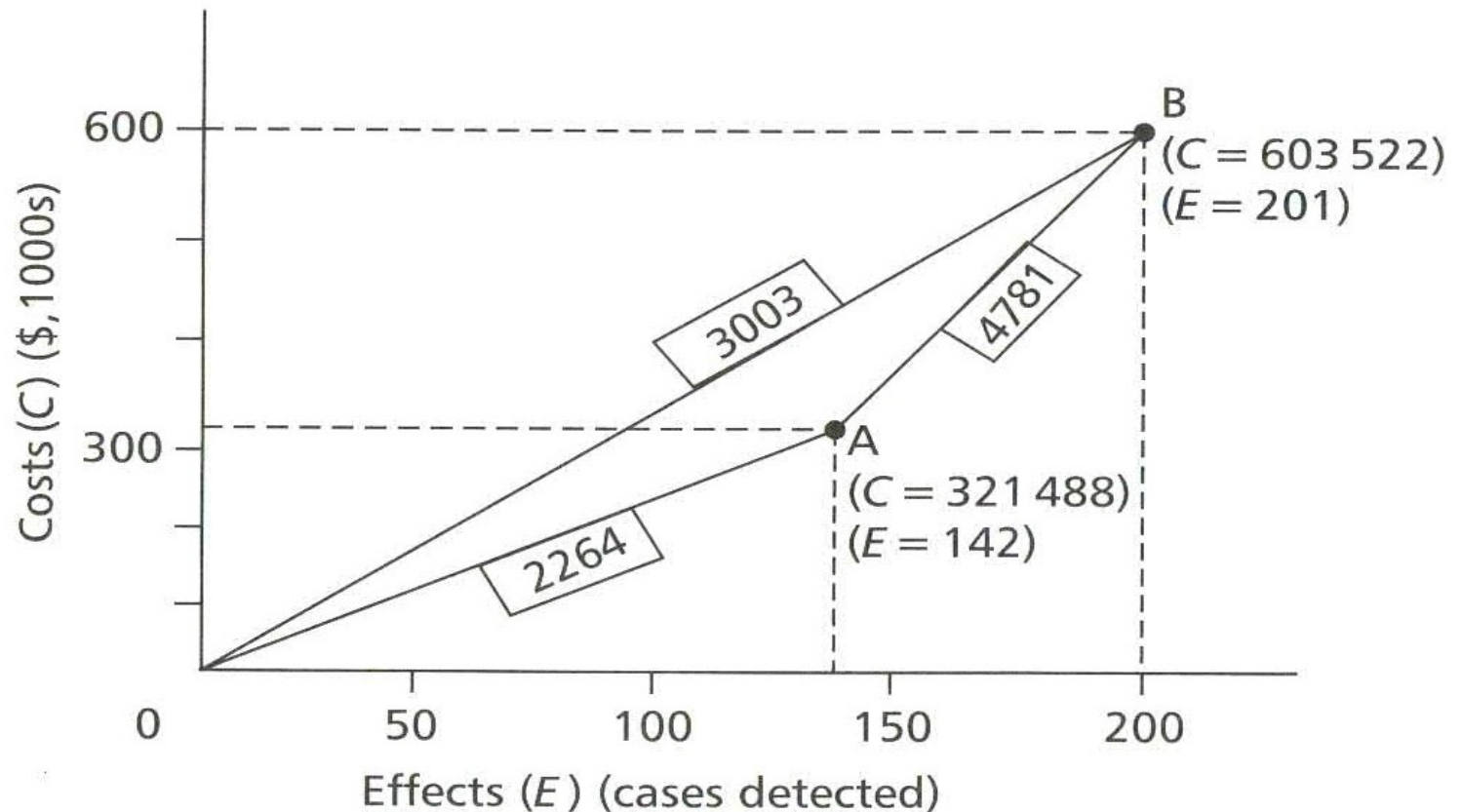


Fig. 3.2 Average and incremental cost-effectiveness ratios.

Example 1

Data inputs	tPA	SK	Increment	Discounted increment
Health care costs	\$24,990	\$24,575	\$415	\$395
Thrombolytic drug cost	\$2,750	\$320	\$2,430	\$2,314.60
Total 1 year costs	\$27,740	\$24,895	\$2,845	\$2,709.60
Life expectancy	15.41	15.27	0.14	0.0829
Incremental CER			?	?

Example with multiple options: illustration of dominance

Option	Cost	Life Expectancy
1	\$750	1
2	\$1,500	1.75
3	\$3,500	2
4	\$3,500	3
5	\$6,000	4
6	\$9,000	4

Example cont'

Option 3& 6 are dominated

Option	Cost	Life Expectancy
1	\$750	1
2	\$1,500	1.75
3	\$3,500	2
4	\$3,500	3
5	\$6,000	4
6	\$9,000	4

Example cont'

Option	Cost	LE (YOLS)	ICER
1	\$750	1	
2	\$1,500	1.75	2 vs 1
3	\$3,500	2	
4	\$3,500	3	4 vs. 2
5	\$6,000	4	5 vs. 4
6	\$9,000	4	

7. Assess the Effect of Uncertainty on the Cost-Effectiveness Ratio

1. Identify uncertain parameters (inputs)

Sources of uncertainty:

- a) Imprecision in measures of resource usage or health effects, e.g., due to sampling variability or use of expert estimates
- b) Uncertainty or instability of unit costs
- c) Methodological controversy e.g., discount rate

Analysis of Uncertainty (cont'd)

2. Assess impact of important sources of uncertainty on results (i.e., the ICER)

Sensitivity Analysis ("What if" analysis) change the value of the "uncertain" parameter by a plausible amount and recalculate the cost-effectiveness ratio

- Most and least conservative estimates
- Confidence interval limits (for sampled data)
- Recommendations in literature, e.g., discount rate

Summary of Uncertainty Analysis

Assesses the impact of changes in values of inputs to the CE analysis on the ICER

1. Conventional sensitivity analysis (deterministic)

- One way or multi-way
- Discrete values or continuous (threshold analysis)

2. Statistical approach

- Generate multiple random samples of total costs and effects and hence ICERs to build a 95% confidence **interval** around the ICER
- Use different techniques for patient-level data vs. model inputs (parameters)

8. Interpret the Cost-Effectiveness Results

1. Is the magnitude of the cost-effectiveness ratio in the base case sufficient to recommended adoption of the technology?

Grades of Recommendation (Laupacis et al. 1992)	
Compelling evidence for adoption	dominates other alternatives
Strong evidence for adoption	incremental ratio < \$20,000 per QALY
Moderate evidence for adoption	incremental ratio \$20,000-100,000/QALY
Weak evidence for adoption	incremental ratio > \$100,000 per QALY
Compelling evidence for rejection	dominated by another alternative

Other Thresholds for the ICER

- USA - \$50,000 per QALY
- UK (NICE) - £20,000-30,000 per QALY
- Problem is that setting a threshold for the ICER is too controversial

Interpreting Cost-Effectiveness Results

2. How strong is the CE ratio in uncertainty analyses (sensitivity and statistical)?
3. What are the study limitations? and were they addressed in the uncertainty analyses?
4. Does the ratio differ significantly across subgroups of patients?
5. What are the major cost drivers?

Examples of Incremental Cost Utility Ratios (\$ per QALY)

Empirical use of Omeprazole in adults with dyspepsia vs. check H. Pylori titer and eradicate if +ve	\$780,000
Pneumococcal vaccine for people > age 65	Cost saving
Warfarin vs. ASA in 65 y.o. with A-fib and medium risk for stroke	\$11,000
Airbags + safety belts vs. belts alone for car driver	\$24,000
Lap/shoulder belts vs. no restraint for rear center car seat	\$2,400,000

Cost-Effectiveness of 11 CVD Prevention Programs (Circulation 2008)

- Smoking cessation was the only prevention strategy that was cost saving over 30 yrs.
- Greatest benefits were from:
 - ASA to high risk individuals \$2,779/QALY gained
 - Controlling pre-diabetes \$17,478/QALY gained
 - Weight reduction in the obese \$18,941/QALY gained
 - Lowering BP in diabetics \$25,317/QALY gained
 - Lowering LDL in CAD \$39,130/QALY gained
- Lowest value from LDL control in low risk people \$270,000/QALY

Other Decision Criteria

- Budget impact
- Maximize health benefits
- Who benefits, e.g., disadvantaged groups
- Increased access to drugs
- Political

Study Questions for Conducting the Analysis and Interpreting the Results

1. What sources of uncertainty might be present in a pharmacoeconomic study that are not present in a clinical drug trial?
2. If one alternative for treating disease A had an incremental cost-utility ratio of \$25,000 per QALY and the other alternative had an incremental ratio of \$32,000 per QALY, which one should be adopted?